MANISH SATISH PAYAPRAPP

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LinkedIn GitHub Belagavi, Karnataka

EDUCATION

Vellore Institute of Technology, Bhopal 2023 – 2027
B.Tech in Computer Science **CGPA: 7.96/10**

Karnataka School Examination and Assessment Board 2023
Class XII **80.66%**

Karnataka School Examination and Assessment Board 2021
Class X **85.44%**

TECHNICAL SKILLS

Languages: JavaScript, C++, Java, Python
Backend Development: Node.js, Express.js, REST APIs, JWT Authentication, Middleware
Frontend Development: React.js, HTML5, CSS3
Databases: MongoDB, MySQL
Security: Authentication and Authorization, bcrypt, Input Validation, OWASP Fundamentals
Tools and Cloud: Git, GitHub, Linux, AWS

PROJECTS

Backend API and Authentication System *Node.js, Express.js, MongoDB, JWT*

- Developed backend functionality using Node.js and Express.js to handle application routes, requests, and business logic.
- Implemented JWT-based authentication and bcrypt password hashing to support secure user authentication.
- Integrated MongoDB for application data management and middleware for protected routes and access control.

Soybean Plant Disease Analysis – Deep Learning Review *Deep Learning, CNN, ResNet, YOLO*

- Reviewed deep learning approaches for soybean disease detection, including CNNs, ResNet, YOLO, transfer learning, and Vision Transformers.
- Compared model performance, computational complexity, inference speed, and deployment feasibility for precision agriculture applications.
- Proposed a deployment-oriented architecture integrating image acquisition, edge preprocessing, disease detection, and cloud analytics.

SecureImage – Cryptography and Image Steganography *Python, NumPy, Tkinter*

- Developed a desktop application combining AES-256-CBC encryption with least-significant-bit image steganography.
- Achieved 3.3 MB embedding capacity in 4K images at 50 dB PSNR and validated 100% data integrity over 100,000 test cycles.

Greedy/Selfish Attack Detection in IEEE 802.11 Networks *Python, Wireshark, PyShark, ML*

- Analyzed wireless traffic using transmission patterns, packet-drop rates, and throughput variations to identify greedy and selfish network behavior.
- Applied Random Forest and K-Means for attack detection and behavioral analysis, achieving 95% detection accuracy with Random Forest and a 92% F1-score with K-Means.

Microsoft Web and Identity Security Research *Burp Suite, OAuth/OIDC*

- Examined authentication workflows, OAuth/OIDC parameters, tokens, redirects, scopes, state, and nonce.
- Tested authorization boundaries using Burp Suite Repeater and HTTP History to assess potential access-control weaknesses.

LEADERSHIP & EXPERIENCE

Event Management Co-Lead – Fusion Club, VIT Bhopal

- Coordinated student teams, event logistics, and execution, strengthening teamwork, communication, and problem-solving.

CERTIFICATIONS & TRAINING

- The Bits and Bytes of Computer Networking – Google
- AWS Technical Essentials
- Blockchain and its Applications – NPTEL, May 2025
- ServiceNow Certified Virtual Internship – SmartBridge Indonesia, June 2026
- Advanced Software Engineering and Dev Internship – MPOnline Limited, May–July 2026
- AI/ML Internship – MPOnline Limited, May–July 2026
- Introduction to Cybersecurity – Cisco Networking Academy